IND-ACC-76 Optical Transport Failure SOP for Indiranagar Exchange Arista Access Node

Node: IND-ACC-76 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_3369

Description

The Indiranagar Exchange Arista Access Node (Role: Access, Node ID: IND-ACC-76) is susceptible to Optical Transport Failures due to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, resulting in alarm ALARM_3369 of severity Minor. This failure domain occurs due to the inherent design of Arista's 400G ZR+ SFP48 transceiver modules, which utilize a combination of coherent and non-coherent detection techniques to achieve high-speed transmission over long distances. The 400G ZR+ module's optical transmitter (Tx) and receiver (Rx) components are vulnerable to degradation and power loss, leading to reduced signal quality and ultimately, network outages. Furthermore, the Indiranagar Exchange's fiber infrastructure may be prone to physical damage, such as cuts or bends, which can compromise signal integrity and trigger the ALARM_3369 alarm.

Pre-requisite Checks

- check: Verify the Indiranagar Exchange's power and environmental status to ensure that the site is operational and within acceptable temperature and humidity ranges.

- check: Confirm that the redundant link status is active and functional to prevent network outages during the troubleshooting process.

- check: Dispatch remote hands to the Indiranagar Exchange to access the node's console port and initiate the troubleshooting procedure.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts or physical damage	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

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Command: show interfaces 1/1/0 detail, Expected Output: Interface 1/1/0, Hardware information: Port-channels, MTU: 9216, Speed: 40000mbps, Flow control: RX, Description: This command displays detailed information about the 1/1/0 interface, including its operational status, MTU, and speed, to help identify any potential issues with the interface.

Command: show log | include ALARM_3369, Expected Output: ALARM_3369: Minor Optical Transport Failure on interface 1/1/0, Description: This command searches the system log for the ALARM_3369 message and
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displays the relevant entry, which helps confirm the failure type and interface affected.

Command: show interfaces 1/1/0 transceiver detail, Expected Output: Transceiver information for interface 1/1/0: SFP48 400G ZR+, Optical transmitter (Tx) and receiver (Rx) status: Degraded, Description: This command displays detailed information about the transceiver module, including its type, optical transmitter and receiver status, and temperature.
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Resolution Steps

Step 1: Step: Replace the transceiver module, Command: replace transceiver 1/1/0, Description: This step involves replacing the faulty transceiver module with a new one to resolve the optical transport failure.

Step 2: Step: Verify the fiber connection, Command: show interfaces 1/1/0 detail, Description: This step involves verifying the physical connection of the fiber cable to the interface to ensure that it is secure and not damaged.

Step 3: Step: Adjust the transceiver module's temperature settings, Command: set transceiver 1/1/0 temperature 45, Description: This step involves adjusting the temperature settings of the transceiver module to a suitable value to prevent degradation and improve signal quality.

Post-Check Verification

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Command: show interfaces 1/1/0 detail, Expected Output: Interface 1/1/0, Hardware information: Port-channels, MTU: 9216, Speed: 40000mbps, Flow control: RX, Description: This command displays detailed information about the 1/1/0 interface, including its operational status, MTU, and speed, to verify that the issue has been resolved.

Command: show log | include ALARM_3369, Expected Output: , Description: This command searches the system log for the ALARM_3369 message and displays the relevant entry to verify that the alarm has been cleared.
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Roll-back Procedures

- Step: Restore the original transceiver module, Command: restore transceiver 1/1/0, Description: This step involves restoring the original transceiver module to its previous state in case the replacement fails to resolve the issue.

- Step: Revert the temperature settings, Command: set transceiver 1/1/0 temperature 40, Description: This step involves reverting the transceiver module's temperature settings to their previous value in case the adjustment fails to resolve the issue.